

Tommaso Mencattini

MSC STUDENT AT EPFL · RESEARCH INTERN AT ISTA, ETH ZURICH, AND GLADIA · PAPERS AT ICLR, ICML, ACL, NEURIPS, ECAI

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Education

École Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, Switzerland

M.S. IN DATA SCIENCE (GPA: 5.7/6)

2024 – Present

- Courses focused on the mathematics of data science, from measure-theoretic probability to reinforcement learning theory.

Vrije Universiteit Amsterdam

Amsterdam, Netherlands

B.S. IN ARTIFICIAL INTELLIGENCE (MAJOR) AND MATHEMATICS (MINOR) (GPA: 9.2/10)

2021 – 2024

- Ranked among the top 7 AI students in the Netherlands for achieving one of the **highest GPAs in a cohort of nearly 300 students** (KHMW Award).
- **Honours Programme:** 30 additional credits of advanced coursework (e.g., Partial Differential Equations) and research projects.

BlueDot Impact

Remote

SUMMER SCHOOL IN AI SAFETY (3 MONTHS)

Summer 2024

- Studied sparse autoencoder diffing and transfer. **Accepted at UniReps NeurIPS 2024 (Blog Post).**

Work Experience

Rome AI Safety, RAIS

Rome

Co-PI

2026 – Ongoing

- Proposer and co-PI, together with *Donato Crisostomi*, of a 7 people research team funded by **Coefficient Giving with \$80k** to develop disentangled, composable steering vectors for fine-grained behavioral control via white-box interventions in activation space.

Locatello Group, ISTA

Vienna

RESEARCH STUDENT

07/2025 – Ongoing

- Research on causality and multimodal foundation models. Co-first author **ORAL (1.2%) ICLR 2026:** framework (theory + library) for unsupervised causal effect discovery in RCTs (PI: *Francesco Locatello*).

GLADIA, Università La Sapienza

Rome

RESEARCH STUDENT

04/2024 – Ongoing

- Scaled evolutionary model merging of LLMs to consumer GPUs; developed ability-estimation theory via Item Response Theory. Now **50x** faster, uses **2%** of the data, with **85%** performance retained (PI: *Emanuele Rodolà*). Two first-author papers at **ICML 2025, ACL 2025**.
- Co-first author paper to **ICLR 2026:** prove injectivity of LLM hidden representations and present the first exact inversion algorithm (**5 M views** on [Twitter](#) tweetprint).

LRE Lab, ETH Zurich

Zurich

RESEARCH STUDENT

06/2023 – 08/2023

- Applied causal inference to explain and assess LLM performance on math word problems (PI: *Mrinmaya Sachan*).

KAI Lab, Vrije Universiteit

Amsterdam

RESEARCH ASSISTANT

01/2023 – 06/2023

- Mitigated hallucinations in LLMs using knowledge graphs for content planning (PI: *Ilaria Tiddi*).

Honors & Awards

2022 **KHMW Young Talent Incentive Award**, National award presented to the student with the highest GPA in their degree program at a Dutch research university.

Amsterdam

2021 **Intelligent Systems Competition**, Ranked 1st out of 270 VU students in the final competition. Developed an intelligent system to play the trick-taking card game Schnapsen against other students' bots.

Amsterdam

Selected Publications

- **T. Mencattini***, F. Montagna*, F. Locatello. “The Rate-Distortion-Polysemanticity Tradeoff in SAEs”. Under Submission.
- M. Iacobelli, A. R. Minut, **T. Mencattini**, D. Crisostomi, A. Santilli, I. Masi, E. Rodolà. “Multi-objective Evolutionary Merging Enables Efficient Reasoning Models”. Under Submission.
- G. Nikolaou*, **T. Mencattini*** et al. “Language Models are Injective and Hence Invertible”. **ICLR 2026**
- **T. Mencattini***, R. Cadei*, et al. “Exploratory Causal Inference in SAEnc”. **ORAL (1.2%) ICLR 2026**
- **T. Mencattini***, A. R. Minut*, et al. “MERGE³: Efficient Evolutionary Merging on Consumer-grade GPUs”. **ICML 2025**.
- S. Facchiano, G. Strano, D. Crisostomi, I. Tallini, **T. Mencattini** et al. “Activation Patching for Interpretable Steering in Music Generation”. Under Review.
- A. R. Minut*, **T. Mencattini*** et al. “Mergenetic: A Simple Evolutionary Model Merging Library”. **ACL 2025** (System Demonstrations).
- M. Calamo, F. De Luzi, M. Macrì, **T. Mencattini** et al. “CICERO: A GPT2-based writing assistant in e-justice”. **ECAI 2023**.
- T. Kutsyk*, **T. Mencattini*** et al. “Do Sparse Autoencoders (SAEs) transfer across base and finetuned language models?” **NeurIPS 2024** (Workshop).